

Assignment 3 - Part 1 Solutions

Question 1.1

For each typing statement, determine whether it is true.

1. $\{ f: (T1 \rightarrow T2), g: (T1 \rightarrow T2), a: T1 \}$
 $\vdash (f (g a)) : T2$

False.

g can be applied to a , and $(g a)$ has type $T2$. However, f expects an argument of type $T1$. Therefore applying f to $(g a)$ is valid only if $T1 = T2$, which is not guaranteed by the statement.

2. $\{ f: (T1 * T2 \rightarrow T3) \}$
 $\vdash (\text{lambda } (x) (f x y)) : (T2 \rightarrow T3)$

False. The variable y is free and is not typed in the environment. Even if the environment also contained $y: T2$, the lambda parameter x would have to be the first argument of f , so the type would be $(T1 \rightarrow T3)$, not $(T2 \rightarrow T3)$, unless $T1 = T2$.

3. $\{ f: (T1 * T2 \rightarrow T3), y: T2 \}$
 $\vdash (\text{lambda } (x) (f x y)) : (T1 \rightarrow T3)$

True.

Under the assumption $x: T1$, the expression $(f x y)$ has type $T3$. Therefore $(\text{lambda } (x) (f x y))$ has type $(T1 \rightarrow T3)$.

Question 1.2.1

Expression: $(\text{lambda } (f1\ g1\ x1\ y1) (f1\ (g1\ y1\ x1) (g1\ x1\ y1)))$

Pool of fresh type variables:

$T0$ = whole expression
 $Tf = f1$
 $Tg = g1$
 $Tx = x1$
 $Ty = y1$
 $T1 = (f1\ (g1\ y1\ x1) (g1\ x1\ y1))$
 $T2 = (g1\ y1\ x1)$
 $T3 = (g1\ x1\ y1)$

First-stage equations:

$T0 = (Tf * Tg * Tx * Ty \rightarrow T1)$
 $Tf = (T2 * T3 \rightarrow T1)$
 $Tg = (Ty * Tx \rightarrow T2)$
 $Tg = (Tx * Ty \rightarrow T3)$

Unification succeeds. A final substitution can be written as:

$S = \{$
 $Ty \mapsto Tx,$
 $T3 \mapsto T2,$
 $Tg \mapsto (Tx * Tx \rightarrow T2),$
 $Tf \mapsto (T2 * T2 \rightarrow T1),$
 $T0 \mapsto ((T2 * T2 \rightarrow T1) * (Tx * Tx \rightarrow T2) * Tx * Tx \rightarrow T1)$
 $\}$

Principal type:

$((A * A \rightarrow B) * (C * C \rightarrow A) * C * C \rightarrow B)$

Question 1.2.2

Expression: $((\text{lambda } (f1) (\text{lambda } (x1) (f1\ x1\ x1))) (\text{lambda } (y1) y1))$

Pool of fresh type variables:

$T0$ = whole expression
 $T1 = (\text{lambda } (f1) (\text{lambda } (x1) (f1\ x1\ x1)))$

```

Tf = f1
T2 = (lambda (x1) (f1 x1 x1))
Tx = x1
T3 = (f1 x1 x1)
T4 = (lambda (y1) y1)
Ty = y1

```

First-stage equations:

```

T1 = (T4 -> T0)
T1 = (Tf -> T2)
T2 = (Tx -> T3)
Tf = (Tx * Tx -> T3)
T4 = (Ty -> Ty)

```

Unification fails. The generated constraints imply:

```

Tf = T4
Tf = (Tx * Tx -> T3)
T4 = (Ty -> Ty)

```

Therefore the algorithm tries to unify:

```

(Ty -> Ty) = (Tx * Tx -> T3)

```

This is impossible because the left side is a one-argument function type and the right side is a two-argument function type. The error is an arity mismatch between function types.

Question 1.2.3

Expression: ((lambda (f1) (lambda (x1) ((f1 x1) x1))) (lambda (y1) y1))

Pool of fresh type variables:

```

T0 = whole expression
T1 = (lambda (f1) (lambda (x1) ((f1 x1) x1)))
Tf = f1
T2 = (lambda (x1) ((f1 x1) x1))
Tx = x1
T3 = ((f1 x1) x1)
T4 = (f1 x1)
T5 = (lambda (y1) y1)
Ty = y1

```

First-stage equations:

```

T1 = (T5 -> T0)
T1 = (Tf -> T2)
T2 = (Tx -> T3)
Tf = (Tx -> T4)
T4 = (Tx -> T3)
T5 = (Ty -> Ty)

```

Unification fails. The generated constraints imply:

```

Tf = T5 = (Ty -> Ty)
Tf = (Tx -> T4)

```

So $Tx = Ty$ and $T4 = Ty$, hence $T4 = Tx$.

But we also have:

```

T4 = (Tx -> T3)

```

Therefore the algorithm tries to solve:

```

Tx = (Tx -> T3)

```

This fails the occurs-check, because Tx occurs inside the type expression on the right. Equivalently, it would require an infinite type.

Question 1.3

Expression: $(\lambda f) (f (\text{right } (\lambda s) (f (\text{left } s))))$

Pool of fresh type variables:

T_0 = whole expression
 T_f = f
 T_1 = $(f (\text{right } (\lambda s) (f (\text{left } s))))$
 T_2 = $(\text{right } (\lambda s) (f (\text{left } s)))$
 T_3 = $(\lambda s) (f (\text{left } s))$
 T_s = s
 T_4 = $(f (\text{left } s))$
 T_5 = $(\text{left } s)$

L and R are fresh auxiliary variables introduced by the left/right rules.

First-stage equations:

$T_0 = (T_f \rightarrow T_1)$

From the outer application of f :

$T_f = (T_2 \rightarrow T_1)$

From the right special form:

$T_2 = R + T_3$

From the inner lambda:

$T_3 = (T_s \rightarrow T_4)$

From the inner application of f :

$T_f = (T_5 \rightarrow T_4)$

From the left special form:

$T_5 = T_s + L$

Unification succeeds. One final substitution is:

$S = \{$
 $T_1 \mapsto T_4,$
 $R \mapsto T_s,$
 $T_3 \mapsto (T_s \rightarrow T_4),$
 $L \mapsto (T_s \rightarrow T_4),$
 $T_5 \mapsto (T_s + (T_s \rightarrow T_4)),$
 $T_2 \mapsto (T_s + (T_s \rightarrow T_4)),$
 $T_f \mapsto ((T_s + (T_s \rightarrow T_4)) \rightarrow T_4),$
 $T_0 \mapsto (((T_s + (T_s \rightarrow T_4)) \rightarrow T_4) \rightarrow T_4)$
 $\}$

Principal type:

$((A + (A \rightarrow B)) \rightarrow B) \rightarrow B$